FROST: certified Hartree–Fock limits of helium and beryllium to hundreds of significant figures

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We report rigorous upper bounds on the closed-shell Hartree–Fock (HF) energies of helium and beryllium whose first 503 and 501 significant figures are correct,

$$E_{\mathrm{HF}}(\mathrm{He}) = -2.861\,679\,995\,612\,238\,878\,775\,543\,740\,022\,563\,429\,412\,879\,776\,752\,447\,053\,906\,864\ldots\ E_{\mathrm{h}},$$
$$E_{\mathrm{HF}}(\mathrm{Be}) = -14.573\,023\,168\,316\,399\,582\,411\,447\,359\,870\,014\,453\,877\,721\,357\ldots\ E_{\mathrm{h}},$$

against previous records of 66 and 43. The method, which we call FROST (Fock radial orbital shooting with Taylor series), uses no basis set. When every occupied orbital is an s orbital, the HF equations reduce to coupled radial differential equations for the orbitals and for one potential per pair of orbitals. We integrate them with Taylor series of order up to 873 in arbitrary-precision arithmetic, sizing every step from the local radius of convergence. The orbital energies and normalisations are fixed by nested secant shooting to a hard wall, which is the only truncation and whose error decays as $e^{-2\kappa R}$. For beryllium each orbital has its own wall, and the canonical equations are solved without enforcing exact orthogonality, which costs energy far below the wall error. The orbitals are piecewise polynomials with dyadic rational coefficients. An independent program therefore evaluates the exact determinant energy in integer arithmetic, and by the variational principle the result is an upper bound free of rounding error. The orbitals are as accurate as the energies. Their nuclear cusp is exact, and for helium the density at the nucleus and $\langle r^k \rangle$ are obtained to about 500 significant figures (previously 32 to 65), with a pointwise error given in closed form by Whittaker functions. On a laptop, the 503-digit helium calculation takes 7.4 minutes and beryllium to 500 digits 4.0 hours. The same code gives the HF energies of H^- through Ne^{8+} to 100 significant figures in about 15 seconds each. We locate the nearest singularities of the helium HF orbital in the complex plane at $r_* = -1.099 \pm 0.577\,i$. This explains why single-exponent Laguerre-type bases converge subgeometrically and predicts the optimised exponent of the previous record to within 3%. Combined with our FLAKE energy, the correlation energy of helium follows to 58 significant figures.

I. INTRODUCTION

The Hartree–Fock (HF) approximation is the oldest working model of a many-electron atom, yet is still widely used throughout quantum chemistry. In 1928, just two years after Schrödinger's equation, Hartree proposed that each electron moves in the field of the nucleus and the averaged charge of the other electrons, with that field iterated until the orbitals it produces regenerate it [1]. Slater showed that this self-consistent field follows from the variational principle [2] and Fock [3] and Slater [4] then took the trial function to be an antisymmetrised product of orbitals, a determinant [5], which satisfies the Pauli principle and adds an exchange term to Hartree's equations. For atoms these are radial differential equations, and they were first solved by integrating them numerically, step by step, on desk calculators. The calculation of Hartree and his father on beryllium [6] was one of the first to include exchange, and the methods of that period are collected in Hartree's book [7].

A major advancement later came from Roothaan [8] and Hall [9] who turned the HF equations into a matrix eigenvalue problem by expanding the orbitals in a fixed basis. With Gaussian functions, whose many-centre integrals have closed forms [10], this made HF calculations on molecules practical once electronic computers became available, and the basis-set expansion has been the standard route ever since. The HF determinant became the reference state of correlated electronic-structure theory; perturbation theory [11], configuration interaction and coupled cluster [12] all expand about it, and Kohn–Sham density functional theory [13] keeps its single-determinant form, with hybrid functionals using its exchange energy directly [14]. Löwdin defined the correlation energy as what the HF limit misses, $E_{\mathrm{corr}} = E - E_{\mathrm{HF}}$ [15]. A correlation energy is therefore only as accurate as the HF limit it is measured from.

For atoms the radial equations can still be solved without explicitly using a basis. Numerical HF programs, notably that of Froese Fischer [16], together with analytical Slater-type expansions [17, 18], became the references against which basis sets are judged. The closed-shell $1s^2$ ground state of helium is the simplest of these problems and is a key focus of the work presented here. It has a single radial orbital, has been solved by almost every method devised for atomic structure, and its HF limit is the standard test of numerical HF methods and basis sets.

Table I shows how the accuracy of the helium HF energy has improved. For more than four decades it stayed at 10–14 significant figures, holding across piecewise polynomials [19], Slater-type expansions [21], B-splines [23], finite elements [25] and our own Laguerre-based treatment in perimetric coordinates [24], a ceiling set largely by double-precision arithmetic. More recently quadruple precision raised it to 25 figures with a pseudospectral method [26] and to 30 with Laguerre-type Λ func-

TABLE I. Published Hartree–Fock energies of the helium ground state (hartree). Values before this work are as compiled in Table 4 of Ref. [29], where each is truncated to its number of correct significant figures (s.f.). Our row shows the first 71 of the 503 correct figures; Appendix A gives all of them.

year	authors	energy	s.f.
1977	Gázquez, Silverstone [19]	−2.86167 99956 122	14
1981	Szalewicz, Monkhorst [20]	−2.86167 99956 12	13
1995	Koga *et al.* [21]	−2.86167 9996	10
2001	Andrae, Brodbeck, Hinze [22]	−2.86167 9996	10
2009	Saito [23]	−2.86167 99956 1	12
2018	King, Baskerville, Cox [24]	−2.86167 99956 12	13
2019	Lehtola [25]	−2.86167 9996	10
2020	Cinal [26]	−2.86167 99956 12238 87877 5544	25
2020	Hatano, Yamamoto [27]	−2.86167 99956 12238 87877 55437 4002	30
2025	Hatano, Yamamoto [29]	−2.86167 99956 12238 87877 55437 40022 56342 94128 79776 75244 70539 06864 69743	66
2026	This work	−2.86167 99956 12238 87877 55437 40022 56342 94128 79776 75244 70539 06864 69743 41883 . . .	503

tions [27]. Hatano and Yamamoto then moved the Λ-function calculation to multiple precision [29]. They used 150 functions with a single optimised exponent and integrals accurate to 80 digits, and reached 66 significant figures, the record before this work.

Basis-set methods face two obstacles at this level, the number of two-electron integrals grows as N^4, and every one must be computed beyond the target precision. More fundamentally, we show in Sec. V that the HF orbital has complex singularities close to the origin. With a single exponential scale these limit convergence to $\exp(-cN^{2/3})$, so the basis must grow rapidly as the target accuracy rises. Extrapolating the record calculation to 500 digits would need about 2 800 functions.

We therefore abandon the basis set and return to Hartree's starting point, the radial differential equations themselves, now integrated to arbitrary order in arbitrary precision. We call the resulting method FROST (Fock radial orbital shooting with Taylor series). The contributions of this paper are:

1. The $1s^2$ HF problem is posed as two coupled ordinary differential equations (ODEs) with two shooting parameters. Separating the electrostatic potential at the nucleus from the orbital energy removes an exact degeneracy, and the two matching conditions then decouple into nested scalar root-finding problems (Sec. II).

2. The ODEs are integrated by arbitrary-order Taylor series in MPFR arithmetic [31]. Taylor methods are established for ODEs [32–34]; here every step is sized from the series' own radius of convergence, and the Frobenius series at the nucleus builds in Kato's cusp condition [35] exactly. The resulting piecewise polynomials are local Taylor expansions of the solution, fixed by the HF equations, not a basis whose coefficients are optimised (Sec. III).

3. The orbital is converted exactly into an admissible trial function whose HF energy is evaluated in integer arithmetic. This gives a variational upper bound with no rounding error, checked by a program independent of the solver (Sec. IV). As far as we know, the closest earlier work is the interval-arithmetic bounding of rounding errors in basis-set HF calculations by Janes and Rendell [36].

4. The complex singularities of the orbital are located and used to explain the convergence of single-exponent expansions (Sec. V).

5. The method extends to several s orbitals (Sec. VII) with one wall per orbital, canonical orbitals certified by a determinant energy that does not assume orthogonality, and nested shooting from the innermost orbital outward. Beryllium is certified to 501 significant figures, against 43 before (Table V).

6. The orbitals are as accurate as the energies (Sec. VIII). The nuclear cusp is exact, the pointwise error follows a closed form in Whittaker functions, and $\langle r^k \rangle$ loses only about two figures per power of r.

Results for helium, the isoelectronic series from H^- to Ne^{8+}, and the correlation energy of helium are in Sec. VI, beryllium is in Sec. VII, and orbital properties are in Sec. VIII. The code is a few hundred lines of Python requiring only 10 seconds on a single laptop core (Apple M4 Pro) to produce 120 digits of accuracy [37].

II. FORMULATION

A. Closed-shell HF for two electrons

Both electrons occupy the orbital $\phi(\boldsymbol{r}) = f(r)/\sqrt{4\pi}$. In atomic units, for an unnormalised radial function f,

the closed-shell HF energy is

$$E[f] = \frac{2(T+V)}{N} + \frac{J}{N^2}, \tag{1}$$

with

$$N = \int_0^\infty f^2 r^2\, dr, \qquad T = \frac{1}{2}\int_0^\infty f'^2 r^2\, dr,$$
$$V = -Z\int_0^\infty f^2 r\, dr, \qquad J = 2\int_0^\infty f^2\, r\, Q(r)\, dr, \tag{2}$$

where $Q(r) = \int_0^r f^2 s^2\, ds$ and $J = (\phi\phi|\phi\phi)$ is the Coulomb self-repulsion of the orbital density. E_{HF} is the infimum of $E[f]$. For $1s^2$ the exchange term cancels half of the Coulomb term, so the Fock operator is $-\frac{1}{2}\nabla^2 - Z/r + V_{\mathrm{H}}$, where V_{H} is the Hartree potential of one electron. Lieb and Simon proved that a minimiser exists when the number of electrons is less than $Z+1$ [38], which here means $Z > 1$; H^- is discussed in Sec. VI C.

With $Y(r) = r\,V_{\mathrm{H}}(r)$ and ϕ normalised, the HF equation and Poisson's equation become

$$rf'' + 2f' = 2\,(Y - Z - \varepsilon r)\, f, \tag{3}$$
$$Y'' = -rf^2. \tag{4}$$

The boundary conditions are that f is regular at the origin and decays, with $Y(0) = 0$, $Y \to 1$ (Gauss's law) and $Y' \to 0$. The total energy is $E = 2\varepsilon - J$.

B. Removing a degeneracy

Eq. (4) does not involve $b = Y'(0) = V_{\mathrm{H}}(0)$, so we write $Y = b\,r + \tilde{Y}$ with $\tilde{Y}(0) = \tilde{Y}'(0) = 0$. Then $Y - \varepsilon r = \tilde{Y} - \mu r$ with $\mu = \varepsilon - b$, and the orbital depends on b and ε only through their difference:

$$rf'' + 2f' = 2\,(\tilde{Y} - Z - \mu r)\, f, \qquad \tilde{Y}'' = -rf^2. \tag{5}$$

In a parametrisation by $(f(0), b, \varepsilon)$ the effects of b and ε on the orbital cancel exactly, which makes the shooting Jacobian ill-conditioned. We use $x = (a, \mu)$ with $a = f(0)$ instead. Integrating Eq. (5) outward from $r = 0$ for given x, the two conditions to be met are

$$f(R) = 0, \qquad N(R) \equiv \tilde{Y}(R) - R\,\tilde{Y}'(R) = 1. \tag{6}$$

The second is the normalisation, because $\frac{d}{dr}(\tilde{Y} - r\tilde{Y}') = r^2 f^2$. The remaining conditions $Y'(R) = 0$ and $Y(R) = 1$ then fix $b = -\tilde{Y}'(R)$, and $\varepsilon = \mu + b$.

C. The wall

We impose $f(R) = 0$ at a finite radius R (a Dirichlet wall) and extend f by zero beyond it. The minimiser on $[0, R]$ is the exact HF orbital of a confined atom. Because the confined orbitals are a subset of the admissible ones, its energy satisfies $E_R \geq E_{\mathrm{HF}}$. The error is set by the squared amplitude of the free orbital at R. In the residual Coulomb field $-(Z-1)/r$ that amplitude behaves as $f \sim r^{\sigma-1}e^{-\kappa r}$ with $\kappa = \sqrt{-2\varepsilon}$ and $\sigma = (Z-1)/\kappa$, so

$$E_R - E_{\mathrm{HF}} \simeq C\, R^{2\sigma} e^{-2\kappa R}. \tag{7}$$

For helium $\kappa = 1.354958$ and each bohr of wall adds $2\kappa/\ln 10 = 1.18$ significant figures. Eq. (7) is checked against the computed energies in Sec. VI. For a target of D digits we take $R = (D+8)\ln 10/(2\kappa)$, rounded up to a multiple of $1/64$.

III. NUMERICAL METHOD

A. Taylor recurrences

About a point $r_0 > 0$ we write $f = \sum_k f_k (r - r_0)^k$, and similarly for $\tilde{Y}$. Products in Eq. (5) become Cauchy convolutions, and equating powers gives the explicit recurrences

$$\tilde{Y}_{k+2} = -\frac{r_0 (f^2)_k + (f^2)_{k-1}}{(k+1)(k+2)},$$
$$f_{k+2} = \frac{g_k - (k+1)(k+2) f_{k+1}}{r_0 (k+1)(k+2)}, \tag{8}$$
$$g_k = 2\big[(\tilde{Y} f)_k - Z f_k - \mu(r_0 f_k + f_{k-1})\big].$$

Here $(uv)_k = \sum_{j \leq k} u_j v_{k-j}$. Starting from the values and slopes at r_0, order p costs $O(p^2)$ multiplications. At $r_0 = 0$, a regular singular point, the left side of Eq. (3) contributes $(k+1)(k+2) f_{k+1}$ to the power r^k, so $f_{k+1} = g_k/[(k+1)(k+2)]$. Only $a = f(0)$ is free, the singular solution $\sim 1/r$ is excluded, and the first step of the recurrence is $f_1 = -Z f_0$: Kato's cusp condition [35] holds exactly. The integrals $\tilde{J} = \int r f^2 \tilde{Y}\, dr$ and T are carried as two further series, $\tilde{J}' = r f^2 \tilde{Y}$ and $T' = \frac{1}{2} r^2 f'^2$. The energy $E = 2\varepsilon - (\tilde{J} + bN)$ and the virial value $-2T$ therefore come out of the same pass.

The orbital is thus represented by polynomials, but it is not expanded in a basis, hence our basis free claim. In a basis-set calculation, including finite elements and B-splines, the coefficients of fixed functions are the unknowns. They are found from integrals and a matrix eigenproblem, and the accuracy is limited by how well those functions can represent the orbital. Here no coefficient is an unknown. Given a and μ, Eq. (8) fixes all of them, and the polynomial on each step is the local Taylor expansion of the solution itself, truncated where its remainder falls below the working precision. Like any high-order integrator of an ODE, the method produces a piecewise polynomial, and Sec. IV uses that function as a trial function. Nothing is optimised over it, however, and its representation error is not what limits the result.

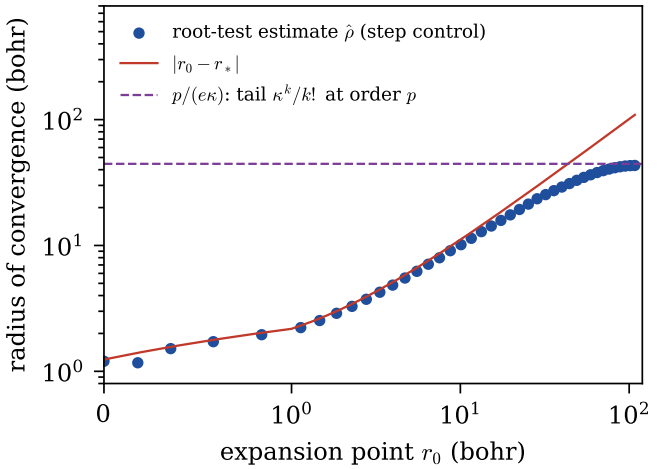


FIG. 1. Radius of convergence of the local Taylor series of the helium orbital (run A of Table II). Points: the root-test estimate used for step control. Red line: the distance to the nearest singularities, located in Sec. V. Dashed line: beyond about 20 bohr a finite order $p = 164$ sees the coefficients $\kappa^k/k!$ of the exponential tail first, which caps the estimate near $p/(e\kappa) = 45$ bohr.

B. Step size

Each step is a fixed fraction c of the radius of convergence, estimated by the root test on the last nine coefficients, $\hat{\rho} = \min_{p-8\le k\le p} |f_0/f_k|^{1/k}$. The step $h = c\hat{\rho}$ is rounded down to a multiple of 2^{-16}, so that every knot is an exact dyadic rational. Truncating the series then costs a relative error of about c^p, and we take $p = \lceil (d+5)/\log_{10}(1/c) \rceil$ for d target digits. The radius grows linearly with distance from the nucleus (Fig. 1), because it is the distance to a fixed pair of complex singularities. The steps therefore grow geometrically, and 51 steps reach $R = 432$ bohr. We use $c = 0.15$ by default; $c = 0.25$ takes fewer steps at higher order.

C. Shooting

For fixed a, $f(R)$ depends almost linearly on μ, because μ sets the amplitude of the solution growing as $e^{\kappa r}$. A secant iteration on μ (inner loop) therefore converges in two or three integrations. The normalisation residual $N(a) - 1$, with the inner solve done for each a, is then smooth, and an outer secant iteration on a solves it.

The natural alternative, a joint 2×2 Newton iteration on Eq. (6), fails at large R. Away from the solution N contains the growing mode squared, a term $\propto e^{2\kappa R}$, so its Jacobian row becomes parallel to that of $f(R)$. At $R = 277$ in a 500-digit run the two rows were parallel to 31 digits, and the basin of the joint iteration shrinks as $e^{-\kappa R}$. The nested scalar solves never leave the growing mode in N, after the first few continuation stages, which take up to 60 integrations from the hydrogenic start, every stage takes 15–23, independent of R.

The wall is reached by continuation as we start from a screened hydrogenic orbital with $\zeta = Z - 5/16$ and $R_0 = 7/\zeta$, and multiply R by 1.2 per stage on a 1/64 grid. Each stage targets the $d_k = \min(D, 2\kappa R_k/\ln 10) + 10$ digits that its wall can deliver, with working precision $d_k + 40$ digits, and updates κ from the current ε. The mesh is frozen within a stage, so both residuals are smooth functions of x. The secant tolerance is 10^{-d_k}, with initial relative offsets of $10^{-(d_k+40)/2}$.

IV. CERTIFICATION

The shooting energy is accurate but not rigorous, being affected by rounding, series truncation and the secant tolerance. The certificate removes all three. It is a separate program of about 100 lines that shares no code with the solver, and it proceeds as follows.

1. It reads the elements $(r_j, h_j, \{f_{jk}\})$ written by the solver. All of these are dyadic rationals, since MPFR numbers are dyadic and the knots lie on a 2^{-16} grid. It verifies exactly that the elements tile $[0, R]$ ($r_0 = 0$, $h_j > 0$, $r_j + h_j = r_{j+1}$).

2. On each element it substitutes $r = r_j + h_j s$, $s \in [0, 1]$, and multiplies by a common power of two so that every polynomial $P_j(s)$ has integer coefficients. The functional (1) is invariant under this scaling.

3. It imposes continuity exactly, by setting $P_j(0)$ equal to $P_{j-1}(1)$, and the node $f(R) = 0$, by adjusting the linear coefficient of the last element. Both adjustments are of the size of the solver's residuals, $\sim 10^{-d}$. With $f = 0$ beyond R, the trial function is a continuous piecewise polynomial with finite kinetic energy and therefore admissible.

4. It evaluates N, T, V and J of Eq. (2) exactly. On each element every integrand is an integer polynomial in s, integrated over $[0, 1]$ with the common denominator $\mathrm{lcm}(1, \ldots, n)$. The inner integral $Q(r)$ is accumulated across elements as an exact polynomial antiderivative.

5. It forms $E[f]$ as a rational number and prints it rounded towards $+\infty$.

The variational principle then gives $E_{\mathrm{HF}} \le E_R \le E[f]$ with no arithmetic error. Any piecewise polynomial would produce a valid bound; the solver only makes the bound tight. The cost is dominated by products of integer polynomials of degree $2p$ with coefficients of a few thousand bits: 16 s at 120 digits and 34 min at 500. The current version computes these products by Kronecker substitution [30]: each polynomial is packed into one integer, so that GMP performs a single multiplication. It returns identical rationals 4–5 times faster at 100–200 digits; the helium timings quoted here predate it. Sec. VII extends the certificate to several orbitals.

V. ANALYTIC STRUCTURE OF THE ORBITAL

A. Complex singularities

The hydrogenic orbital e^{-Zr} is an entire function whilst the HF orbital is not: Eqs. (3)–(4) are nonlinear, and their solutions have movable singularities. Running B's Frobenius series at $r = 0$, of order 626, shows sign patterns that rotate with k, which is the signature of a complex-conjugate pair. We fit the two-term recurrence $f_{k+1} = Af_k + Bf_{k-1}$ to its last 120 coefficients; the roots of $w^2 - Aw - B$ are the reciprocals of the singular points. This gives

$$r_* = -1.099 \pm 0.577\,i, \qquad |r_*| = 1.241 \text{ bohr}. \tag{9}$$

The value changes by 7×10^{-4} between coefficient windows. Fits about $r_0 = 0.6$, 1.5, 3.5, 8 and 17 bohr locate the same pair to within 3×10^{-2}. Figure 1 confirms that the radius of convergence is $|r_0 - r_*|$ along the whole inner region, which is what makes the Taylor steps grow geometrically.

B. Consequences for single-exponent bases

The Λ functions of Refs. [27, 29] are $L_n^{(2)}(2\zeta r)e^{-\zeta r}$, and our earlier basis [24] is of the same type. Expanding f in them amounts to expanding $g(x) = f(x/2\zeta)\,e^{x/2}$ in Laguerre polynomials and the following two features of f limit how fast the coefficients decay.

- **Tail mismatch.** The tail $e^{-\kappa r}$ becomes $e^{-\beta x}$ with $\beta = (\kappa - \zeta)/2\zeta$. The generating function $\sum_n L_n^{(\alpha)}(x)t^n = (1-t)^{-\alpha-1}e^{-xt/(1-t)}$ shows that its coefficients fall as t^n, with $|t| = (\zeta - \kappa)/(\zeta + \kappa)$.

- **Singularity.** The singular point maps to $x_* = 2\zeta r_*$. A Laguerre series converges inside the largest parabola $\mathrm{Re}\sqrt{-x} < \tau$ free of singularities, with coefficients decaying as $e^{-2\tau\sqrt{n}}$ [39]. Here $\tau = \sqrt{2\zeta}\,\mathrm{Re}\sqrt{-r_*} = 1.0815\sqrt{2\zeta}$.

The energy error is quadratic in the orbital error, so the number of correct digits is about $(2/\ln 10)\min\bigl[N \ln|t|^{-1},\, 2\tau\sqrt{N}\bigr]$. Raising ζ pushes the singularity outward in x but worsens the tail mismatch. The best exponent balances the two, giving $\zeta_{\mathrm{opt}} \propto N^{1/3}$ and an error $\exp(-cN^{2/3})$, slower than any geometric rate.

This leading-order estimate, which ignores prefactors, can be tested against the record calculation. For $N = 150$ it gives $\zeta_{\mathrm{opt}} = 4.98$, against the numerically optimised $\zeta = 5.1448$ of Ref. [29]. At their exponent the two mechanisms limit the energy to 70 and 74 digits respectively, so the optimised exponent indeed sits where they balance. The 66 digits observed are within the neglected prefactors. On the same estimate, 503 digits would need $N \approx 2\,800$ with $\zeta \approx 13$, or about 10^{13} two-electron integrals at 500-digit precision which goes to show how efficient the FROST method is.

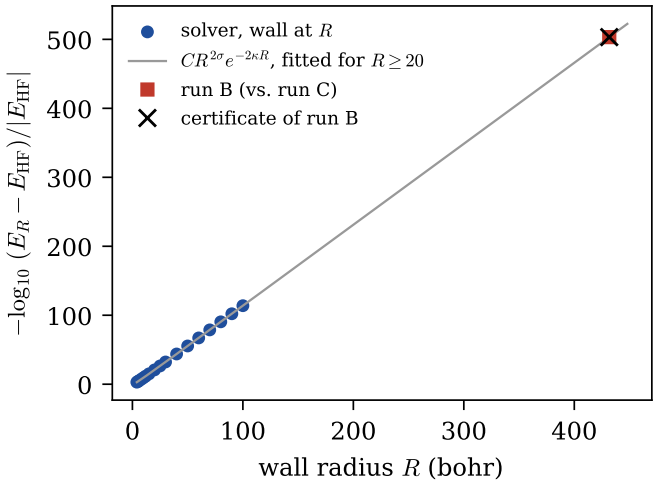


FIG. 2. Accuracy of the confined energy E_R against wall radius. Points up to $R = 100$ are solver energies. The line is Eq. (7) with C fitted for $R \geq 20$; it predicts run B's error at $R = 432$ to 1%. Square: run B against run C. Cross: the certified bound from run B.

VI. RESULTS

A. Helium

Table II lists the helium runs. Run B (target 500 digits) and run C differ in wall radius, Taylor order, step fraction and working precision. The certificate of run B gives

$$\begin{aligned} E_{\mathrm{HF}} \leq -2.861\,679\,995\,612\,238\,878\,775 \\ 543\,740\,022\,563\,429\,412\,879 \\ 776\,752\,447\,053\,906\,864\,697\ldots\ E_{\mathrm{h}}, \end{aligned} \tag{10}$$

with the digits in full in Appendix A. The error analysis follows.

- **The bound.** Eq. (10) is rigorous. It lies -2.6×10^{-509} from run B's own energy, so the adjustments of Sec. IV are invisible at this precision.

- **Its distance from E_{HF}.** Run B lies 1.70×10^{-503} above run C. Eq. (7), fitted only to walls up to 100 bohr (Fig. 2), predicts 1.7×10^{-503} for run B's wall alone. The whole difference is therefore run B's wall. The same law gives 3×10^{-519} for run C's wall, and run C's truncation and rounding are set at 510 and 540 digits.

- **Conclusion.** Hence $E[f] - E_{\mathrm{HF}} = 1.70 \times 10^{-503}$ to within about 10^{-509}. That is a relative error of $10^{-503.2}$, and the first 503 significant figures of Eq. (10) are those of E_{HF}.

TABLE II. Helium runs. D: target digits; R: wall radius (bohr); p: Taylor order; c: step as a fraction of the radius of convergence; bits: MPFR working precision; steps: Taylor steps to R. Times are for one core of an Apple M4 Pro. The last two columns are $-\log_{10}|E-E_{\rm HF}|/|E_{\rm HF}|$ for the solver energy and for the certified bound, with run C as the reference.

run	D	R	p	c	bits	steps	solver	certificate	solver s.f.	bound s.f.
A	120	108.77	164	0.15	564	41	11 s	16 s	124.1	124.1
B	500	431.66	626	0.15	1827	51	7.4 min	34 min	503.2	503.2
C	500	445.00	856	0.25	1793	30	8.2 min	–	reference	–

As internal checks, run B satisfies the virial theorem to $|E+2T| = 2.0\times10^{-500}$. This is 1168 times its wall error, close to the ratio $2\kappa R = 1170$ expected for the pressure term that a wall adds to the virial theorem. All 66 figures of Ref. [29] agree with Eq. (10). The orbital energy is $\varepsilon = -0.917955562856339510 1\ldots$; Table III gives more digits.

B. The isoelectronic series

Only Z changes along the series. Table III lists certified 100-digit results for $Z = 1$–10, each taking 10–16 s plus a certificate of similar cost. Every exact bound lies above its independent reference and agrees with it to 103–108 significant figures. Because κ grows with Z, the wall needed for 100 digits shrinks from 409 bohr for H^- to 13 bohr for Ne^{8+}.

C. H^-

H^- is often described as unbound in HF theory. This means that $E_{\rm HF}(H^-)$ lies 0.0120702656 $E_{\rm h}$ above the energy of H($1s$) plus a free electron, $-\frac{1}{2}$ $E_{\rm h}$, so HF predicts a negative electron affinity. The closed-shell problem itself still has a normalisable solution. At large r the Hartree potential $Y/r \to 1/r$ cancels the nuclear attraction exactly, so the orbital sees a short-range potential deep enough to bind it: $\varepsilon = -0.0462224456276869354\ldots$ and $f \sim e^{-\kappa r}/r$ with $\kappa = 0.3040$ ($\sigma = 0$ in Eq. (7)).

The calculation shows this directly. The orbital energy converges to 15 digits by $R \approx 76$ bohr and does not change as the wall moves out to 409 bohr, whereas an orbital held in place only by the wall would have $\varepsilon \to 0$ as R grows. The H^- case lies just outside the existence theorem of Lieb and Simon, which needs fewer than $Z+1$ electrons [38]. The certified value bounds the infimum of the closed-shell functional whether or not a minimiser exists. The closed-shell constraint is what keeps the ion together. Without it, a determinant with one electron in H($1s$) and the other spread out to infinity approaches $-\frac{1}{2}$ $E_{\rm h}$, below the restricted energy. Because κ is small, each significant figure costs 3.8 bohr of wall, against 0.85 bohr for helium.

D. Correlation energy of helium

Our FLAKE calculation [40] gives an upper bound on the nonrelativistic ground-state energy of helium, and estimates from convergence laws that the exact energy lies less than 3.3×10^{-60} below it (60 significant figures). Subtracting $E_{\rm HF}$ at either end of that interval gives the same correlation energy,

$$\begin{aligned} E_{\rm corr} = -0.042\,044\,381\,421\,880\,719\,535 \\ 615\,505\,171\,841\,017\,284\,045 \\ 534\,364\,843\,285\,135\,48\ldots\ E_{\rm h}, \end{aligned} \tag{11}$$

correct to 58 significant figures. Its accuracy is now set entirely by the correlated calculation.

E. Computational cost

All runs used Python with `gmpy2` (MPFR and GMP) on one core of an Apple M4 Pro.

- **Solver.** Each integration costs $O(p^2)$ multiplications per step at d-digit precision, with $p \propto d$ and $O(\log R)$ steps. There are $O(\log R)$ continuation stages of about 20 integrations each. The solver took 11 s at 120 digits and 7.4 min at 500, scaling as about $D^{2.6}$ over that range.
- **Certificate.** It works with integers of up to a few thousand bits and scales as about $D^{3.4}$: 16 s at 120 digits and 34 min at 500.
- **Memory.** The 500-digit orbital file, which holds every Taylor coefficient, is 18 MB so the memory footprint is very low.

VII. SEVERAL s ORBITALS: BERYLLIUM

A. Equations

For a closed shell of n doubly occupied s orbitals f_i, every pair of orbitals has a potential

$$Y_{ij}(r) = r\int_0^\infty \frac{f_i(s)f_j(s)\,s^2}{\max(r,s)}\,ds, \qquad Y_{ij}'' = -r f_i f_j, \tag{12}$$

TABLE III. Closed-shell HF energies $E_{\rm HF}$ and orbital energies ε of the two-electron atoms (hartree), truncated. R is the wall radius (bohr) of the certified 100-digit run. "agree" is $-\log_{10}$ of the relative difference between that run's exact certified bound and an independent solver run with a larger wall, $c = 0.25$ and different working precision. Appendix A gives the bounds rounded up to 100 significant figures.

Z	ion	$E_{\rm HF}$	ε	R	agree
1	H^-	−0.48792 97343 70851 74869 28383 72756 80759 06985	−0.04622 24456 27686 93542 6028	409.0	107.8
2	He	−2.86167 99956 12238 87877 55437 40022 56342 94128	−0.91795 55628 56339 51014 7216	91.8	104.2
3	Li^+	−7.23641 52014 52256 24768 29074 24015 50788 80999	−2.79236 44022 45808 08521 9025	52.6	103.7
4	Be^{2+}	−13.61129 94306 19109 03495 58301 80936 76999 6618	−5.66711 55889 28776 00495 431	36.9	103.5
5	B^{3+}	−21.98623 44668 24343 01967 70457 83810 15678 7231	−9.54197 84830 02732 90264 850	28.5	103.3
6	C^{4+}	−32.36119 28757 18012 33131 52026 75639 74699 5556	−14.41689 16445 67255 25006 614	23.2	103.3
7	N^{5+}	−44.73616 39649 42473 12978 20380 85154 85704 6668	−20.29183 17112 25191 51808 053	19.5	103.3
8	O^{6+}	−59.11114 27019 13653 85859 84716 50258 65974 1750	−27.16678 78561 29001 32879 335	16.9	103.2
9	F^{7+}	−75.48612 64062 60509 83759 56212 09428 90145 1559	−35.04175 43745 26538 66787 855	14.9	103.2
10	Ne^{8+}	−93.86111 35192 31338 15497 81638 81035 30642 9111	−43.91672 79750 63963 31759 665	13.3	103.2

with $Y_{ij}(0) = 0$ and $Y'_{ij}(R) = 0$. Y_{ii} is the Coulomb potential of orbital i and Y_{ij} $(i \neq j)$ the exchange potential between i and j. For s orbitals only the monopole term appears, so in this form exchange is a multiplication, not an integral operator. The canonical HF equations generalise Eq. (3) to

$$r f_i'' + 2 f_i' = 2\Big[\Big(\sum_j 2Y_{jj} - Y_{ii} - Z - \varepsilon_i r\Big) f_i - \sum_{j\neq i} Y_{ij} f_j\Big]. \tag{13}$$

As for helium, $Y_{ij} = b_{ij} r + \tilde{Y}_{ij}$, so the orbitals depend on ε_i and b_{ij} only through $\mu_i = \varepsilon_i - \sum_j 2b_{jj} + b_{ii}$ and $\nu_{ij} = b_{ij}$ $(i \neq j)$. The Taylor recurrences of Sec. III carry over term by term, and the Frobenius series again gives $f_i'(0) = -Z f_i(0)$. Beryllium $(n = 2)$ has two orbitals, three pair potentials and five shooting parameters, $(a_1, a_2, \mu_1, \mu_2, \nu_{12})$ with $a_i = f_i(0)$. With $n = 1$ the same code reproduces the helium results.

B. One wall per orbital

The $1s$ orbital decays as $e^{-\kappa_1 r}$ with $\kappa_1 = 3.077$, the $2s$ as $e^{-\kappa_2 r}$ with $\kappa_2 = 0.786$. A single wall placed for the $2s$ would make the $1s$ growing mode $e^{\kappa_1 r}$ demand $\kappa_1/\kappa_2 \approx 4$ times the working precision. Each orbital therefore has its own wall R_i, chosen so that $e^{-2\kappa_i R_i}$ is the same for all, and is zero beyond it; for the run of Eq. (15) below, $R_1 = 190.1$ and $R_2 = 743.7$ bohr. These orbitals are still admissible, so the variational bound is unaffected.

C. Canonical orbitals and a certificate without orthogonality

With a common wall, canonical orbitals of one Fock operator are orthogonal automatically. With unequal walls they are not, and the constrained optimum needs a Lagrange multiplier, a term $\lambda_{12} r f_j$ in each equation. Enforcing exact orthogonality with it proved badly conditioned. The overlap S_{12} responds to λ_{12} only at the 10^{-4} level, and a Newton step large enough to correct it forces a $1s$ tail far above its natural size $e^{-\kappa_1 R_1}$, which the exchange term passes into the $2s$ growing mode. We keep the canonical equations instead ($\lambda_{12} = 0$, so $\nu_{12} = b_{12}$ is fixed by $Y'_{12}(R) = 0$) and accept the overlap that results: $|S_{12}| = 4.9 \times 10^{-315}$ for the run of Eq. (15). Because the energy is stationary, the cost is of order S_{12}^2, far below the wall error.

The certificate makes this rigorous. For any orbitals, orthonormal or not, the closed-shell determinant energy is

$$E = 2\sum_{ij} P_{ij} h_{ij} + \sum_{ijkl} P_{ij} P_{kl} \big[2(ij|kl) - (ik|jl)\big], \tag{14}$$

with $P = S^{-1}$. It depends only on the span of the orbitals. For piecewise polynomials every overlap S_{ij}, one-electron integral h_{ij} and two-electron integral $(ij|kl) = \int \rho_{ij} r Q_{kl}\, dr + \int \rho_{kl} r Q_{ij}\, dr$ (with $\rho_{ij} = f_i f_j$ and $Q_{kl}(r) = \int_0^r \rho_{kl} s^2 ds$) is a rational number, computed as in Sec. IV, and S^{-1} is inverted exactly. For $n = 1$, Eq. (14) reduces to Eq. (1), and the generalised certifier returns the identical rational for helium.

D. Shooting

Two further changes were needed.

- **Nested solves from the innermost orbital outward.** Solving $f_1(R_1) = 0$ and $f_2(R_2) = 0$ jointly fails. While μ_1 is wrong, the $1s$ orbital carries a growing mode that the exchange term feeds into the $2s$ equation, and $f_2(R_2)$ acquires spurious roots in μ_2. Every trial μ_2 therefore re-solves μ_1

first. The $1s$ rounding error that remains is amplified near R_1 and passed on through exchange, which limits μ_2 to about $10^{-0.6d}$. The energy, quadratic in orbital errors, is unaffected, so the scalar solves stop when the residual reaches its noise floor. The outer unknowns (a_1, a_2, ν_{12}) are found by Broyden's method from a fresh finite-difference Jacobian at each wall radius; a Jacobian carried over from the previous radius converges only linearly.

- **Homotopy start.** Screened hydrogenic orbitals are too crude a start for the $2s$. We scale the electron–electron terms by t and follow the solution from $t = 0$, where the orbitals are hydrogenic and exact, to $t = 1$ in 40 steps at low precision with a linear predictor.

E. Results

Table IV lists the runs. The certificate of run A500 gives

$$E_{\mathrm{HF}}(\mathrm{Be}) \le -14.573\,023\,168\,316\,399\,582\,411\,447\,359\,870\,014\,453\,877\,721\,357\,774\,005\,413\ldots\ E_{\mathrm{h}}, \tag{15}$$

with all digits in Appendix A. It lies 7.08×10^{-501} above run B500. Eq. (7) for the $2s$ wall, with $\sigma = 1/\kappa_2$ in the residual field $-1/r$ and C fitted to run A100 alone, predicts 7.03×10^{-501} for run A500's wall. The difference is therefore the wall error of the certified run, and its relative size, $10^{-501.3}$, makes the first 501 significant figures of Eq. (15) those of $E_{\mathrm{HF}}(\mathrm{Be})$. All 43 figures of Ref. [29] agree (Table V). The orbital energies are $\varepsilon_{1s} = -4.732669897455726203 96\ldots$ and $\varepsilon_{2s} = -0.309269551573442817 60\ldots$. The run took 4.0 hours and its certificate 13 minutes.

VIII. QUALITY OF THE ORBITALS

The energy is often a forgiving test for method accuracy as it is stationary, so an orbital error of relative size δ changes it only at order δ^2, whereas every other property changes at order δ. In a basis-set calculation, properties therefore carry far fewer correct figures than the energy, and fewest near the nucleus. At the 66-figure energy of Ref. [29], the cusp ratio of helium has 30 correct figures, $\rho(0)$ 32 and $\langle r^{-2}\rangle$ 35 (Table VI). FROST orbitals behave differently, because their only error is the wall and that error is known in closed form. We evaluate properties of the same exact trial orbitals that are certified, by exact integration of the piecewise polynomials.

A. The nucleus

The first step of the Frobenius recurrence is $f_1 = -Zf_0$ (Sec. III). For $Z = 2$ and $Z = 4$ it involves no rounding, so the cusp ratio $f'(0)/f(0)$ of every certified orbital, both beryllium orbitals included, is exactly the rational number $-Z$, and so is $\rho'(0)/2\rho(0)$. The higher Frobenius coefficients also come from the HF equation itself, so the orbital has the analytic form of the HF solution at the nucleus, not merely the right cusp. The properties dominated by this region are then as accurate as any other. Runs B and C give the same $\rho(0)$ to 500 significant figures and the same $\langle r^{-2}\rangle$ to 500.

B. Pointwise error

Once the density has decayed, $Y = 1$ up to terms of order $e^{-2\kappa r}$, and Eq. (3) becomes the Coulomb equation for charge $Z - 1$ at energy ε. Its decaying and growing solutions are $W_{\sigma,1/2}(2\kappa r)/r$ and $M_{\sigma,1/2}(2\kappa r)/r$, where W and M are Whittaker functions [41]. The free orbital is $A\,W_{\sigma,1/2}(2\kappa r)/r$. A wall at R adds the multiple of the growing solution that makes f vanish there, so the relative error of the confined orbital is

$$\frac{f_R}{f} - 1 \simeq -\frac{W_{\sigma,\frac12}(2\kappa R)\,M_{\sigma,\frac12}(2\kappa r)}{M_{\sigma,\frac12}(2\kappa R)\,W_{\sigma,\frac12}(2\kappa r)} \propto \Big(\frac{R}{r}\Big)^{2\sigma} e^{-2\kappa(R-r)}, \tag{16}$$

plus a constant of the order of the energy error from the shifts of ε and of the normalisation. Figure 3 compares Eq. (16) for run B with the measured difference between runs B and C. From $r = 10$ bohr out to the wall they agree to one part in 10^5. Run B's orbital is therefore correct to 500 figures at the nucleus and to about 250 at $r = R/2$, and its error is known everywhere.

The same figure tests the tail itself. Run C, divided by $A\,W_{\sigma,1/2}(2\kappa r)/r$ with A fitted at $r = 222$ bohr, approaches 1 as $e^{-2\kappa r}$ while the exponentially small part of the Hartree potential dies away. Near $r = R/2$, where this error meets the growing wall error, the agreement reaches about 261 significant figures. The decay constant, the power $r^{\sigma-1}$ and every correction to it are thus those of the exact Coulomb tail. The asymptotic normalisation of the helium HF orbital, with $\int f^2 r^2\,dr = 1$, is $A = 1.390801162856952920 83\ldots$

C. Expectation values

Because $M_{\sigma,1/2}W_{\sigma,1/2}$ tends to a constant, Eq. (16) changes f^2r^2 by the same amount u at every r in the Coulomb region. Then $\delta\langle r^k\rangle \approx u\,[R^{k+1}/(k+1) - R\langle r^k\rangle]$, and each power of r costs about $\log_{10} R - \log_{10}(\langle r^{k+1}\rangle/\langle r^k\rangle) \approx 2$ significant figures. With u fitted at $k = 1$, this reproduces the differences between runs B and C for every k from 1 to 12 to within 0.04%, and even $\langle r^{12}\rangle$ is correct to 474 figures. Table VI lists the

TABLE IV. Beryllium runs. Run A_D targets D digits and is certified; run B_D is an independent check with larger walls, $c = 0.25$ and hence a different Taylor order and mesh. R_1, R_2: walls of the $1s$ and $2s$ orbitals (bohr); other columns as in Table II, with run B500 as the reference. Runs were made up to four at a time on the same laptop.

run	D	R_1	R_2	p	c	bits	steps	solver	certificate	solver s.f.	bound s.f.
A100	100	40.4	158.1	140	0.15	498	54	170 s	18 s	103.0	103.0
B100	110	44.2	172.8	208	0.25	498	32	289 s	–	112.9	–
A200	200	77.8	304.5	261	0.15	830	59	23 min	86 s	202.3	202.3
B200	210	81.6	319.1	374	0.25	830	34	20 min	–	212.2	–
A500	500	190.1	743.7	626	0.15	1827	65	239 min	13 min	501.3	501.3
B500	510	193.9	758.3	873	0.25	1827	38	232 min	–	reference	–

TABLE V. Published Hartree–Fock energies of the beryllium ground state (hartree), as compiled in Table 4 of Ref. [29] and truncated there to their significant figures; our row shows the first 66 of 501 correct figures.

year	authors	energy	s.f.
1995	Koga *et al.* [21]	−14.57302 317	10
2001	Andrae, Brodbeck, Hinze [22]	−14.57302 317	10
2009	Saito [23]	−14.57302 31683	12
2019	Lehtola [25]	−14.57302 3168	11
2020	Cinal [26]	−14.57302 31683 16399 58241 145	25
2022	Hatano, Yamamoto [28]	−14.57302 31683 16399 58241 14473 599	30
2025	Hatano, Yamamoto [29]	−14.57302 31683 16399 58241 14473 59870 01445 38777 2	43
2026	This work	−14.57302 31683 16399 58241 14473 59870 01445 38777 21357 77400 54138 05222 7220...	501

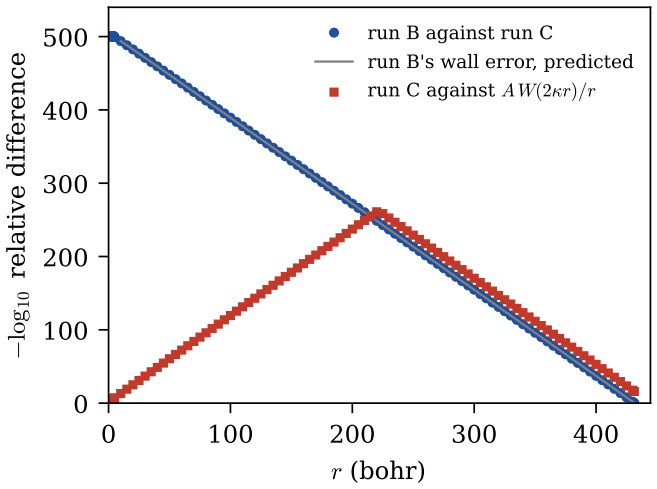


FIG. 3. Pointwise accuracy of the 500-digit helium orbital. Circles: relative difference between runs B ($R = 431.7$) and C ($R = 445$). Line: run B's wall error predicted by Eq. (16). Squares: run C against the exact Coulomb tail $A\,W_{\sigma,1/2}(2\kappa r)/r$, limited at small r by the non-Coulomb part of the potential and at large r by the wall.

results beside those of Ref. [29]. The virial ratio of run B deviates from -2 only by its wall pressure (Sec. VI), at the 10^{-500} level. For beryllium we take T from each orbital normalised on its own; the kinetic energy of the $P = S^{-1}$ determinant, like any property, is good only to about $|S_{12}|$.

D. Beryllium

Both beryllium cusp ratios are exactly -4, against 19 correct figures in Ref. [29]. Runs A500 and B500 agree on $\rho(0)$ to 498 figures and on the $\langle r^k \rangle$ of the density to 478–498, the wall error of run A500. The moments of the individual $1s$ and $2s$ orbitals agree only to 315–321 figures, comparable to $|S_{12}| = 4.9 \times 10^{-315}$. This is consistent with a rotation of order S_{12} within the occupied space, which leaves the determinant and its density unchanged to first order. Physical properties depend only on the density, and the per-electron averages of Table VI are density properties.

IX. DISCUSSION

For closed-shell atoms whose occupied orbitals are all s orbitals, the HF problem is one-dimensional, and it can be solved to essentially any precision in time polynomial in the number of digits. This needs no basis set, no two-electron integrals and no extrapolation. The only approximation is the wall, whose error decays exponentially and follows a law (Eq. (7)) that is verified inde-

TABLE VI. Orbital properties per electron, compared with the multiple-precision Λ-function results of Ref. [29] (its Tables 4 and 6). CC is the cusp ratio $\rho'(0)/2\rho(0)$ (exactly $-Z$), VR the virial ratio V/T (exactly -2), $\rho(0)$ the total density at the nucleus, and $\langle r^k\rangle$ is in bohr^k. Values are truncated to 22 figures. "here" is the number of correct significant figures, from the relative difference between our two independent runs (He: runs B and C; Be: runs A500 and B500), and for VR from the virial ratio of run B (He) or A500 (Be). "Ref." is the number reported in Ref. [29]; every figure printed there agrees with ours, up to rounding of the last.

	He			Be		
property	value	here	Ref.	value	here	Ref.
CC	-2 (exact)	exact	30	-4 (exact)	exact	19
VR	-2	500	68	-2	498	55
$\rho(0)$	3.59591 82637 62248 92070 6...	500	32	35.38771 67410 93786 44935...	498	20
$\langle r^{-2}\rangle$	5.99550 36673 12524 59727 3...	500	35	14.40450 64454 41573 89932...	498	23
$\langle r^{-1}\rangle$	1.68728 22152 81009 40400 8...	500	65	2.10219 94129 30678 79400 7...	498	41
$\langle r\rangle$	0.92727 34047 31493 19741 89...	497	63	1.53220 39722 81112 53930 2...	495	39
$\langle r^2\rangle$	1.18482 84799 09482 92632 5...	495	61	4.32969 28432 10096 87398 1...	492	38
$\langle r^3\rangle$	1.94063 62073 75593 29252 7...	493	60	15.78772 44502 52942 44512...	490	37
$\langle r^4\rangle$	3.88794 66255 72633 62739 1...	491	58	67.66413 23859 49548 84927...	488	35
$\langle r^5\rangle$	9.22119 73006 38698 13804 8...	488	57	331.37292 59118 13439 5126...	486	34
$\langle r^6\rangle$	25.27839 04132 93347 64158...	486	56	1826.40384 40394 25393 288...	484	33
$\langle r^7\rangle$	78.65249 19190 10120 85779...	484	54	11199.59304 93608 21497 21...	482	32
$\langle r^8\rangle$	273.84506 80901 48372 3311...	482	53	75677.78895 46800 29670 81...	480	31
$\langle r^9\rangle$	1054.83264 42597 84461 806...	480	52	558916.88019 33108 60490 3...	478	30

pendently. The solver and the certificate play separate roles. The solver is free to use any arithmetic or heuristic, because the certificate turns its output into a bound that depends on nothing but exact integer arithmetic and the variational principle. The same construction applies whenever an energy functional is an integral of polynomials in the trial function: a piecewise polynomial with dyadic coefficients always admits an exact evaluation.

Beryllium shows that the method is not specific to two electrons. Each pair of occupied orbitals adds a Poisson equation and each orbital adds a shooting parameter. The problems this creates, an inner orbital's growing mode and the coupling through exchange, were solved by giving each orbital its own wall, by solving from the innermost orbital outward, and by a certificate that does not need orthogonality. For p and d shells, exchange brings multipoles $r_<^k/r_>^{k+1}$ with $k \geq 1$. Their integrals over a polynomial element produce logarithms of knot ratios, so the certificate would need rigorous interval enclosures of those logarithms instead of exact rationals; the solver is unchanged. Open shells such as lithium need the restricted open-shell energy expression. The Be-like ions follow from Sec. VII by changing Z.

The main limitation is that the bound is one-sided. The distance to E_{HF} is established by an independent run and by the measured wall law, not by a rigorous lower bound. Producing a certified lower bound for a nonlinear eigenproblem at this precision remains open.

DATA AND CODE AVAILABILITY

The solvers (`oracle.py` for one orbital, `oracle_ns.py` for several), the certificate (`certify.py`), orbital properties (`properties.py`), a test, the 100-digit isoelectronic values and the script generating every table and figure of this paper are in Ref. [37].

Appendix A: Digits

The certified bound for helium from run B, truncated to its 503 correct significant figures (truncation keeps every value here an upper bound):

$$E_{\mathrm{HF}}(\mathrm{He}) \le -2.8616799956\,1223887877\,5543740022\,5634294128\,7977675244\,7053906864\,6974341883\,6262795569\,6043994965\,3409977869\,0984175165\,7319012702\,3919647238\,6125667693\,7899153381\,1865805702\,9646754000\,6262349883\,4061443950\,9601437492\,4135358476\,7851824834\,0821214097\,0971618545\,3258894677\,1779765923\,3024533037\,8073662227\,1074836399\,7217257536\,6709739632\,5040135317\,3639891864\,9058074245\,0858489821\,0511928945\,7211993802\,4044762317\,9624760205\,8266331292\,1264856378\,6408084536\,2955571466\,4474164321\,9393621412\,5615947217\,4331793223\,0647798571\,8919570787\,0010781657\,47\ldots$$

The certified bound for beryllium from run A500, truncated to its 501 correct significant figures:

$$E_{\mathrm{HF}}(\mathrm{Be}) \le -14.5730231683\,1639958241\,1447359870\,0144538777\,2135777400\,5413805222\,7220687300\,0193822635\,1455080316\,9376936498\,9813411646\,0590414220\,8100746275\,1047907294\,5277805073\,8975228184\,9318054044\,5506276329\,3865670096\,0809617754\,8999876512\,9672202879\,1450889866\,6849101213\,4508707507\,9427000347\,0399581021\,8875529001\,2123902027\,0945596918\,4619912969\,1544989534\,9196332042\,5651173890\,8727366296\,9171855704\,7823504902\,4927158415\,9882076703\,9342895408\,1793383926\,2557682238\,9455941180\,4164488745\,9691458140\,6525412122\,2562597650\,3205668895\,2590379295\,553928939\ldots$$

The certified bounds for the two-electron isoelectronic series, rounded up to 100 significant figures:

Z	ion	E_{HF} (hartree)
1	H^-	−0.4879297343 7085174869 2838372756 8075906985 3910785709 6269038281 1340665609 7897185876 8966814715 5846566757
2	He	−2.8616799956 1223887877 5543740022 5634294128 7977675244 7053906864 6974341883 6262795569 6043994965 340997786
3	Li^+	−7.2364152014 5225624768 2907424015 5078880999 6155485795 0494422097 9096948083 9824349926 8208719101 959449017
4	Be^{2+}	−13.6112994306 1910903495 5830180936 7699966184 6449778750 8490544069 2629053406 9058737132 8045073910 84123264
5	B^{3+}	−21.9862344668 2434301967 7045783810 1567872310 6164671870 4966978607 1405795390 7507459393 2780368884 77734442
6	C^{4+}	−32.3611928757 1801233131 5202675639 7469955566 8383224096 7460246834 1345694182 6975127407 2131126585 40991441
7	N^{5+}	−44.7361639649 4247312978 2038085154 8570466688 6590871341 6298429652 2643541074 4982503068 5743194168 15832539
8	O^{6+}	−59.1111427019 1365385859 8471650258 6597417502 6580481271 9121555099 4989409315 4057585162 9328732427 93226341
9	F^{7+}	−75.4861264062 6050983759 5621209428 9014515596 0041484019 1470841147 8116942143 1274221352 6242319785 50956336
10	Ne^{8+}	−93.8611135192 3133815497 8163881035 3064291112 8543638382 1944783379 8796142782 0812917287 1166295321 85129503

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